Alex Towell
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(618) 977-1746

Alex Towell

PhD Student | AI Researcher | Mathematician

About Me PhD student in Computer Science at Southern Illinois University, specializing in AI safety, alignment, and cybersecurity. Leading ransomware prevention research achieving 10^{-8} false-negative rates. Architecting production systems for Illinois DOT while maintaining 100+ open-source projects in AI, cryptography, and mathematics. Seeking to contribute to cutting-edge AI research with real-world impact.

Education

2024 - Present, Southern Illinois University
PhD in Computer Science

- Research Focus: Cybersecurity, AI Safety, Alignment, Machine Learning
- Advisor: Dr. Hiroshi Fujinoki
- Expected Graduation: 2028

2021 - 2023, Southern Illinois University-Edwardsville
MS in Mathematics

- GPA: 3.9/4.0
- Concentration: Statistics and Operations Research
- Awarded R.N. Pendergrass Graduate Scholarship

2012 - 2015, Southern Illinois University-Edwardsville
MS in Computer Science

- GPA: 4.0/4.0
- Concentration: Artificial Intelligence
- Awarded Outstanding Graduate Student (School of Engineering)

2006 - 2010, Southern Illinois University-Edwardsville
BS in Computer Science

- GPA: 3.6/4.0
- Awarded Outstanding Junior (Dept. of Computer Science)

Technical Skills

Languages
Python, R, C/C++, Java, Scala, JavaScript, TypeScript, Bash

AI/ML
PyTorch, HuggingFace, CUDA, LangChain, Fine-Tuning, Transformers

Systems
Linux, Docker, Git, PostgreSQL, Elasticsearch, Redis, REST APIs

Cryptography
Encrypted Search, Secure Indexes, Oblivious Computing, Bloom Filters

Frameworks
React, Node.js, FastAPI, Flask, Jupyter, LaTeX

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Publications

- H. Fujinoki, **A. Towell**, V.A. Thota, "Preventing Ransomware Damages using In-Operation Off-Site Backup to Achieve a 10^{-8} False-Negative Miss-Detection Rate," *International Conference on Computing and Intelligence (ICCI)*, 2025
- **A. Towell**, "Reliability Estimation in Series Systems: Maximum Likelihood Techniques for Right-Censored and Masked Failure Data," Master's Project, Department of Mathematics and Statistics, Southern Illinois University Edwardsville, 2023
- **A. Towell**, H. Fujinoki, "Estimating How Confidential Encrypted Searches Are Using Moving Average Bootstrap Method," *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2016
- **A. Towell**, "Encrypted Search: Enabling Standard Information Retrieval Techniques Using New Secure Index Types," Master's Thesis, ProQuest Dissertations & Theses, 2015

Selected Research & Projects

AlgoTree

Developed a Python package offering utilities for tree-like data structures, including FlatForest and TreeNode representations. Features a command-line tool `jt` for creating, manipulating, querying, and visualizing trees, akin to `jq` but tailored for hierarchical data.

Elasticsearch-LM

Fine-tuning smaller Large Language Models (LLMs) for translating natural language queries into structured Elasticsearch Query DSL. Emphasizes synthetic data generation to enhance model training, aiming to democratize sophisticated API interactions.

Reverse-Process Synthetic Data Generation

Developed novel algorithms generating synthetic data via reverse-process methods to facilitate LLM-based mathematical reasoning and theorem proving. Focus on creating high-quality training data for complex reasoning tasks.

Oblivious Computation

Researching cryptographic methods enabling secure computation trade-offs between space and security without full homomorphism. Applications in privacy-preserving cloud computing.

IDOT D8 Infrastructure System

Architected and implemented comprehensive full-stack application for traffic monitoring and infrastructure management. Large-scale production system handling real-time data processing for Illinois transportation.

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Academic Experience

2024–Present, Research Assistant, SIU

- Research in cybersecurity under Dr. Hiroshi Fujinoki, focusing on ransomware detection and prevention using probabilistic data structures and backup strategies
- Lead developer for Illinois Department of Transportation District 8 (IDOT D8) full-stack application
- Developed large-scale codebase for traffic data processing and infrastructure management systems
- Co-authored paper on achieving 10^{-8} false-negative miss-detection rates in ransomware prevention (ICCI 2025)

2021–2022, Teaching Assistant, SIUE

- Taught lab courses in mathematics and statistics

2012–2015, Research/Graduate/Teaching Assistant, SIUE

- Supported research in AI and machine learning while pursuing MS in Computer Science
- Taught undergraduate computer science lab sessions
- Developed encrypted search algorithms for confidential data retrieval systems